Problem 1

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Problem 1 (a) *Suppose $f'(2)$ exists. Which of the following must be true?*

(i) $\lim_{x \to 2} f(x)$ *must exist, but* $\lim_{x \to 2} f(x) \neq f(2)$

(ii) $\lim_{x \to 2} f(x) = f(2)$.

(iii) $\lim_{x \to 2} f(x) = f'(2)$

(iv) $\lim_{x \to 2} f(x)$ *need not exist.*

(b) *Assuming that* $\lim_{x \to 0} \frac{\sin(x)}{x} = 1$, *which of the following is true?*

(i) $\frac{0}{0} = 1$

(ii) *the tangent line to* $y = \sin(x)$ *at* $(0, 0)$ *has slope* 1.

(iii) *you can cancel the* x*'s.*

(iv) *for all* x *near* 0, $\sin(x) = x$.

(v) *for all* x *near* 0, $\sin(x) \approx x$.